

Theory of Computation

500+ One-Liners for Rapid Revision

IBPS SO (IT Officer) Examination - Complete Preparation Series

100% Syllabus Coverage - MCQ-Oriented - Zero Filler

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1. Foundations and Preliminaries

Alphabets, Strings, Operations

- An **alphabet (Sigma)** is defined by exactly **2** properties: it is **finite** and **non-empty**.
- There are **2** symbols in the standard binary alphabet, namely **0** and **1**.
- A **string** has exactly **1** key length measure, written $|w|$, counting its **symbols**.
- The empty string has length **0** and is written **epsilon** (or **lambda**).
- Concatenation satisfies **1** length law: $|xy| = |x| + |y|$.
- Concatenation has exactly **1** property it lacks: it is **not commutative** (so $ab \neq ba$).
- The **identity** element for concatenation is the **1** string **epsilon**, since $w.\text{epsilon} = w$.
- A string of length **n** has exactly **n+1** prefixes, including **epsilon** and the **whole string**.
- A string of length **n** has exactly **n+1** suffixes, by the same counting.
- A string of length **n** has exactly $n(n+1)/2$ non-empty substrings, all of which are **contiguous**.
- A string of length **n** can have up to 2^n subsequences, which need **not** be contiguous.
- There are **2** distinct concepts confused as one: a **substring** (contiguous) and a **subsequence** (skips allowed).
- Reversal obeys **1** key law: $(xy)^R = y^R x^R$ (order flips).
- w^0 equals exactly **1** value: **epsilon** (any string to the power zero).
- There are **3** positional relations of a string: **prefix** (front), **suffix** (back), and **substring** (middle).

Empty String vs Empty Language

- There are **3** confusable objects: the string **epsilon**, the language $\{\text{epsilon}\}$ (size **1**), and the empty language **phi** (size **0**).
- The empty language **phi** contains exactly **0** strings.
- The language $\{\text{epsilon}\}$ contains exactly **1** string, namely **epsilon**.
- Concatenating any language with **phi** gives **1** result: **phi** (the empty language annihilates).
- Concatenating any language with $\{\text{epsilon}\}$ returns exactly **1** thing: the **original language L**.

Kleene Star, Plus, Sigma-star

- The **Kleene star L^*** always contains at least **1** guaranteed string: **epsilon**.
- The **Kleene plus L^+** contains **epsilon** only in **1** case: when **epsilon** is already in **L**.
- There is **1** relation linking them: $L^+ = L.L^*$ (and $L^* = L^+ \cup \{\text{epsilon}\}$).
- **phi**^{*} equals exactly **1** language: $\{\text{epsilon}\}$.
- **Sigma**^{*} is the set of **all** finite strings and is **countably infinite**.
- **Sigma**⁺ differs from **Sigma**^{*} by exactly **1** string: it excludes **epsilon**.
- The number of languages over an alphabet is **uncountably infinite** ($2^{\aleph_0}$), exceeding the **countable Sigma**^{*}.

Chomsky Hierarchy

- The **Chomsky hierarchy** has **4** grammar types: **Type 0**, **Type 1**, **Type 2**, and **Type 3**.
- The hierarchy maps to **4** machines: **TM**, **LBA**, **PDA**, and **Finite Automaton**.
- The hierarchy maps to **4** language classes: **RE**, **Context-Sensitive**, **Context-Free**, and **Regular**.
- **Type 0** grammars are **unrestricted** and correspond to **Turing Machines** / **RE** languages.
- **Type 1** grammars are **context-sensitive** with rule $|\alpha| \leq |\beta|$, recognised by an **LBA**.
- **Type 2** grammars are **context-free** with a **single** variable on the LHS, recognised by a **PDA**.

- **Type 3** grammars are **regular** (**left-** or **right-linear**), recognised by a **Finite Automaton**.
- The Chomsky hierarchy numbering is **inverted**: **Type 0** is the most powerful, **Type 3** the least.
- There are **4** strictly nested classes: **Regular** $\subset$ **CFL** $\subset$ **CSL** $\subset$ **RE**.
- **Noam Chomsky** proposed the hierarchy in **1956**.

2. Finite Automata

DFA Basics

- A **DFA** is defined by a **5-tuple**: $(Q, \Sigma, \delta, q_0, F)$.
- A DFA transition function has exactly **1** target: $\delta: Q \times \Sigma \rightarrow Q$.
- A DFA has exactly **1** start state, namely q_0 .
- A DFA accepts w under exactly **1** condition: $\delta^*(q_0, w) \in F$.
- A DFA accepts **epsilon** in exactly **1** case: when $q_0 \in F$.
- A DFA over k symbols with n states has exactly $n \cdot k$ transition-table entries, all of which must be **defined**.
- A **dead/trap state** has **1** defining behaviour: all its transitions **loop back** to itself and it is **non-accepting**.
- A minimal DFA has at most **1** dead state.
- The language of a DFA is **regular**, the **1** class finite automata recognise.
- There are **3** things a DFA forbids: **multiple** transitions, **epsilon** moves, and **undefined** transitions.

DFA Construction Patterns

- "Length mod $n = r$ " needs exactly n states arranged in a **cycle**.
- "Number of a's divisible by k " needs exactly k states counting **modulo k** .
- "Contains substring w " needs exactly $|w|+1$ states tracking the **longest matched prefix**.
- "Ends with string w " needs $|w|+1$ states using **prefix tracking**.
- "Binary value divisible by k " needs k states tracking the **remainder mod k** , updating as $(2 \cdot r + b) \bmod k$.
- " n -th symbol from the end is 1" needs exactly 2^n states in the minimal DFA but only $n+1$ in an NFA.

NFA

- An **NFA** is a **5-tuple** with **1** difference from a DFA: $\delta: Q \times \Sigma \rightarrow 2^Q$ (power set).
- An NFA from one state on one symbol may go to **0, 1, or many** states.
- An NFA accepts w under **1** condition: at least **1** path ends in F .
- An NFA rejects w only when **all** paths fail.
- **NFA** and **DFA** recognise exactly the **same** class: the **regular** languages.
- Non-determinism gives a finite automaton **0** extra power, only added **convenience**.
- An n -state NFA converts to a DFA with at most 2^n states.

Epsilon-NFA

- An **epsilon-NFA** adds **1** feature: transitions on **epsilon** (no input consumed).
- **epsilon-closure(q)** always contains at least **1** state: q itself.
- There are **3** equivalent finite models: **DFA**, **NFA**, and **epsilon-NFA**.
- Epsilon transitions add **0** extra language power, only **structural** convenience.

Conversions and Blow-up

- The **subset construction** converts an NFA to a DFA with worst-case 2^n states.
- An **epsilon-NFA** converts to a DFA using **epsilon-closure** plus the **subset** method.
- The DFA start state in subset construction is the **1** set **epsilon-closure($\{q_0\})$.**
- A DFA subset is **accepting** under **1** condition: it contains at least **1** NFA **final** state.
- Converting a **DFA to NFA** is **trivial** because every DFA is already an **NFA**.

DFA Minimization

- DFA minimization yields a **unique** minimal DFA, guaranteed by the **Myhill-Nerode** theorem.
- The **table-filling** algorithm runs in $O(n^2)$ and marks **(final, non-final)** pairs first.
- **Hopcroft's** minimization algorithm runs in $O(n \log n)$, the **best known** bound.
- Minimization requires **1** preliminary step: remove **unreachable** states first.
- Two states are **distinguishable** if **some** string makes one **accept** and the other **reject**.
- The number of **Myhill-Nerode classes** equals the number of states in the **minimal DFA**.
- There are **4** steps to table-filling: remove unreachable, mark final/non-final, propagate marks, merge unmarked.

Moore and Mealy Machines

- A **Moore machine** is a **6-tuple** whose output depends on **1** thing: the **current state**.
- A **Mealy machine** is a **6-tuple** whose output depends on **2** things: **state** and **input**.
- A Moore machine emits **$n+1$** outputs for an input of length **n** .
- A Mealy machine emits **n** outputs for an input of length **n** .
- Moore and Mealy are **equivalent** in power, differing by exactly **1** output (the initial-state output).
- A Moore machine places outputs on **states**; a Mealy machine places them on **transitions**.
- Converting Mealy to Moore can need up to **$m * |O|$** states from **m** original states.

3. Regular Expressions

Definition and Precedence

- A **regular expression** has **3** base cases: **phi**, **epsilon**, and a **symbol a**.
- A regular expression has **3** operators: **union (+)**, **concatenation (.)**, and **Kleene star (*)**.
- Operator precedence has **3** levels, highest to lowest: **star**, **concatenation**, **union**.
- **Kleene's theorem** equates **3** things: **regular expressions**, **finite automata**, and **regular languages**.

Algebraic Identities

- Union has **3** basic laws: **commutative**, **associative**, and **idempotent ($r+r=r$)**.
- **phi** plays **2** roles: **identity** for union (**$\text{phi}+r=r$**) and **annihilator** for concatenation (**$\text{phi}.r=\text{phi}$**).
- **epsilon** is the **1** identity for concatenation: **$\text{epsilon}.r = r$** .
- There is **1** star identity: **$(r^*)^* = r^*$** .
- **phi^*** simplifies to **1** expression: **epsilon**.
- There are **2** famous FALSE identities: **$(r+s)^* \neq r^*+s^*$** and **$(rs)^* \neq r^* s^*$** .
- One key unrolling law: **$r^* = \text{epsilon} + r r^*$** .

Conversions

- **Thompson's construction** converts an RE of size **n** to an **epsilon-NFA** with **$O(n)$** states.
- **Arden's theorem** solves **$R = Q + RP$** with the **1** unique solution **$R = QP^*$** .
- Arden's uniqueness needs exactly **1** condition: **epsilon not in P**.
- The **state-elimination** method converts a **finite automaton to an RE** by removing intermediate states.

4. Properties of Regular Languages

Closure Properties

- Regular languages are closed under all **3** Boolean operations: **union**, **intersection**, **complement**.
- Regular languages are closed under **6+** operations including **concatenation**, **star**, and **reversal**.
- Complement closure uses **1** trick: swap **final** and **non-final** states of a **complete DFA**.
- Intersection closure uses the **product construction**, giving at most **$m \cdot n$** states.
- Regular languages form a **Boolean algebra**, closed under **all** Boolean operations.
- Regular languages are closed under **homomorphism** and **inverse homomorphism**, totalling **0** exceptions among common operations.

Pumping Lemma

- The regular pumping lemma splits a string into **3** parts: **x** , **y** , **z** .
- The regular pumping lemma has **3** conditions: **$|y| \geq 1$** , **$|xy| \leq p$** , and **$xy^i z \in L$ for all $i \geq 0$** .
- The pumping lemma proves only **1** thing: **non-regularity** (it is **necessary**, **not sufficient**).
- **$a^n b^n$** is the classic language proven **non-regular** by the pumping lemma.
- The condition **$|xy| \leq p$** forces **y** into the **first p** symbols.

Myhill-Nerode

- The **Myhill-Nerode theorem** is the **1** test that is both **necessary AND sufficient** for regularity.
- **L** is regular iff its Myhill-Nerode relation has a **finite** number of **equivalence classes**.
- The **index** (class count) equals the size of the **minimal DFA**.
- An **infinite** index implies the language is **not regular**.

Decision Properties

- Regular languages have **4** decidable standard problems: **membership**, **emptiness**, **finiteness**, **equivalence**.
- **Membership** for a DFA is decided in **$O(|w|)$** time by **simulation**.
- **Emptiness** is decided by **1** test: whether any **final** state is **reachable**.
- **Finiteness** is decided by checking for a **cycle** on a path to a **final** state.
- **Equivalence** of regular languages is **decidable** (unlike for CFLs).

Regular Grammars

- A **right-linear** grammar uses productions of form **$A \rightarrow aB$** or **$A \rightarrow a$** .
- A **left-linear** grammar uses productions of form **$A \rightarrow Ba$** or **$A \rightarrow a$** .
- **Pure left-linear** and **pure right-linear** grammars both generate exactly the **regular** languages.
- **Mixing** left- and right-linear rules can produce a **non-regular** (context-free) language.

5. Context-Free Grammars

Definition and Derivations

- A **CFG** is a **4-tuple**: (V, T, P, S) .
- A CFG production has exactly **1** variable on the LHS, of form $A \rightarrow \alpha$.
- There are **2** canonical derivation orders: **leftmost (LMD)** and **rightmost (RMD)**.
- A **leftmost derivation** always expands the **leftmost** variable first.
- A **rightmost derivation** always expands the **rightmost** variable first.
- Each parse tree corresponds to exactly **1** leftmost and **1** rightmost derivation.

Parse Trees and Ambiguity

- A **parse tree** has **1** root (S), with **leaves** forming the **yield**.
- A grammar is **ambiguous** if some string has ≥ 2 distinct **parse trees**.
- Ambiguity is a property of the **grammar**, while **inherent ambiguity** is a property of the **language**.
- $\{a^i b^j c^k : i=j \text{ OR } j=k\}$ is a classic **inherently ambiguous** language.
- Deciding if an arbitrary CFG is ambiguous is **undecidable**.

CFG Design

- $\{a^n b^n\}$ is generated by **1** rule set: $S \rightarrow aSb \mid \epsilon$.
- Palindromes over $\{a,b\}$ use $S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$.
- Balanced parentheses use $S \rightarrow SS \mid (S) \mid \epsilon$.
- Equal numbers of a's and b's use $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$.

Simplification

- A **useless symbol** fails **1** of **2** tests: it is **non-generating** or **unreachable**.
- Simplification removes **non-generating** symbols **before unreachable** ones.
- The recommended simplification order has **3** steps: remove **epsilon-productions**, then **unit productions**, then **useless symbols**.
- A **unit production** has **1** form: $A \rightarrow B$ (single variable RHS).
- An **epsilon-production** has **1** form: $A \rightarrow \epsilon$.
- $S \rightarrow \epsilon$ is kept only in **1** case: when **epsilon** is in the language.

Normal Forms

- **Chomsky Normal Form** allows exactly **2** production shapes: $A \rightarrow BC$ and $A \rightarrow a$.
- A CNF derivation of a length- n string uses exactly $2n-1$ steps.
- **Greibach Normal Form** requires each RHS to start with exactly **1** terminal, form $A \rightarrow a \alpha$.
- A GNF derivation of a length- n string uses exactly n steps.

Left Recursion and Factoring

- **Left recursion** $A \rightarrow A \alpha$ breaks **top-down (LL)** parsers.
- Left recursion is removed by rewriting to **right recursion**: $A \rightarrow \beta A'$, $A' \rightarrow \alpha A' \mid \epsilon$.
- **Left factoring** extracts **common prefixes** to enable **predictive** parsing.

- Removing left recursion fixes 0 ambiguity problems (they are independent issues).

6. Pushdown Automata

Definition

- A **PDA** is a **7-tuple**: (**Q**, **Sigma**, **Gamma**, **delta**, **q0**, **Z0**, **F**).
- A PDA equals a finite automaton plus exactly **1** extra memory: a **stack** (**LIFO**, **unbounded**).
- A PDA's stack alphabet is denoted **Gamma**, with initial symbol **Z0**.
- An **instantaneous description** of a PDA has **3** parts: **state**, **remaining input**, **stack contents**.

Acceptance Modes

- A PDA has **2** acceptance modes: by **final state** and by **empty stack**.
- The **2** acceptance modes are **equivalent** for an **NPDA** (both recognise CFLs).

Power and Determinism

- **CFG** and **NPDA** are **equivalent**, both characterising the **context-free** languages.
- A **DPDA** recognises the **DCFLs**, a **proper subset** of the **CFLs**.
- **NPDA** is strictly more powerful than **DPDA** (unlike NFA vs DFA where power is **equal**).
- **{w w^R}** (even palindromes) is a CFL accepted by an **NPDA** but **not** by any **DPDA**.
- A single stack handles exactly **1** counting comparison, which is why **a^n b^n** is a CFL but **a^n b^n c^n** is not.
- **DCFLs** (by final state) are closed under **complement** but not under **union** or **intersection**.

7. Properties of Context-Free Languages

Closure

- CFLs are closed under **4** operations: **union**, **concatenation**, **star**, **reversal**.
- CFLs are NOT closed under **3** operations: **intersection**, **complement**, **difference**.
- **CFL intersect Regular** is always a **CFL** (1 stack survives).
- The non-closure of intersection is shown by $\{a^n b^n c^m\} \cap \{a^m b^n c^n\} = \{a^n b^n c^n\}$, which is **not** a CFL.

Pumping Lemma for CFL

- The CFL pumping lemma splits a string into **5** parts: **u, v, w, x, y**.
- The CFL pumping lemma pumps **2** substrings simultaneously: **v** and **x**.
- The CFL pumping lemma has **2** length conditions: $|vwx| \leq p$ and $|vx| \geq 1$.
- $a^n b^n c^n$ is proven **non-CFL** by the CFL pumping lemma but is **context-sensitive**.

Decision Properties

- CFLs have **3** decidable standard problems: **membership**, **emptiness**, **finiteness**.
- **Membership** for CFLs is decided by the **CYK algorithm** in $O(n^3)$ time using **CNF**.
- CFL **equivalence** is **undecidable**, the key contrast with regular languages.
- At least **4** CFL problems are undecidable: **equivalence**, **ambiguity**, **$L = \Sigma^*$** , and **intersection-emptiness**.

8. Turing Machines

Definition

- A **Turing Machine** is a **7-tuple**: $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$.
- A standard TM transition produces **3** outputs: a **new state**, a **written symbol**, and a **move L or R**.
- A TM tape alphabet **Γ** contains the **input alphabet** plus the **blank B**.
- A TM has exactly **2** head-move directions: **Left** and **Right**.
- A TM configuration captures **3** things: **tape contents**, **head position**, **current state**.

Behaviour and Variants

- A TM can have **3** outcomes on an input: **accept**, **reject**, or **loop forever**.
- There are **5+** TM variants all equal in power: **multi-tape**, **non-deterministic**, **multi-track**, **two-way**, **enumerator**.
- A **k-tape** TM is simulated by a single-tape TM with at most **$O(t^2)$** slowdown.
- A **non-deterministic TM** adds **0** language power over a deterministic TM (only possible time speed-up).
- An **enumerator** shows a language is **recursive** iff it lists strings in **lexicographic** order.

UTM and Church-Turing

- A **Universal Turing Machine** takes **1** encoded input $\langle M, w \rangle$ and **simulates M** on **w**.
- The UTM is the theoretical basis of the **stored-program computer**.
- The **Church-Turing thesis** is a **thesis (not a theorem)** equating **effective computability** with **TM-computability**.
- The thesis is supported by **3** equivalent models: **TMs**, **lambda-calculus**, and **recursive functions**.

9. Recursive and Recursively Enumerable Languages

Definitions

- A **recursive** language has **1** defining machine: a TM that **halts on every input** (a **decider**).
- A **recursively enumerable** language has a TM that **accepts** members but may **loop** on non-members.
- There are **3** synonyms for RE: **semi-decidable**, **Turing-recognisable**, **recursively enumerable**.
- **Recursive** is a **proper subset** of **RE** (every recursive language is RE, not conversely).

Closure

- **Recursive** languages are closed under **complement**; **RE** languages are **not**.
- Both recursive and RE are closed under **4** operations: **union**, **intersection**, **concatenation**, **star**.
- **Post's theorem** states **L** is recursive iff **both L and L'** are **RE**.
- If **L** is RE but not recursive, then **L'** is **not even RE**.

10. Decidability and Undecidability

Decidable Problems

- **Regular** languages have at least **5** decidable problems: **membership**, **emptiness**, **finiteness**, **equivalence**, **universality**.
- **CFLs** have **3** decidable problems: **membership**, **emptiness**, **finiteness**.
- **Equivalence** is decidable for **regular** but **undecidable** for **CFL** and **TM**.

Halting Problem and Reductions

- The **Halting Problem** is **undecidable**, proven by **Alan Turing** in **1936** via **diagonalization**.
- **HALT_TM** is **RE but not recursive**.
- A **mapping reduction** $A \leq_m B$ uses **1** computable function with **x in A iff $f(x)$ in B** .
- If $A \leq_m B$ and **B** is decidable, then **A** is **decidable** (and undecidability flows the other way).

Rice's Theorem and PCP

- **Rice's theorem** states every **non-trivial** property of a TM's **language** is **undecidable**.
- Rice's theorem applies to **semantic** (language) properties, not **syntactic** (machine) ones.
- A **trivial** property (always true or always false) is **decidable**, the **1** exception to Rice.
- The **Post Correspondence Problem (PCP)** is **undecidable** and asks for a matching **sequence of indices**.

RE-but-not-recursive Examples

- There are **3** canonical RE-but-not-recursive languages: **A_{TM}** , **$HALT_{TM}$** , and **L_u** .
- The **complement** of **A_{TM}** is **not even RE**.
- The **diagonal language** **L_d** (machines not accepting their own code) is **not RE**.
- **$\{a^n b^n c^n\}$** is **recursive** (context-sensitive), unlike the undecidable TM languages.

11. Complexity (Awareness)

- **Time complexity** counts **steps** and **space complexity** counts **tape cells**, the **2** TM resources.
- The class **P** contains problems solvable in **polynomial** time by a **deterministic** TM.
- The class **NP** stands for **Non-deterministic Polynomial**, with poly-time **verifiable** certificates.
- **P subset of NP**, and whether **P = NP** is the famous **open Millennium Prize** problem.
- **NP-complete** requires **2** conditions: being **in NP** and being **NP-hard**.
- The first NP-complete problem was **SAT**, by the **Cook-Levin theorem** in **1971**.
- At least **6** classic NP-complete problems exist: **3-SAT**, **Clique**, **Vertex Cover**, **Hamiltonian Cycle**, **TSP**, **Subset-Sum**.
- **NP-hard** problems need **not** be in NP, e.g. the **Halting problem** is NP-hard but **undecidable**.

12. High-Yield Exam Facts (Rapid Fire)

Class vs Machine vs Grammar

- The **Regular** class pairs with **1** machine (**FA**) and **1** grammar (**Type 3**).
- The **CFL** class pairs with **1** machine (**PDA**) and **1** grammar (**Type 2**).
- The **CSL** class pairs with **1** machine (**LBA**) and **1** grammar (**Type 1**).
- The **RE** class pairs with **1** machine (**TM**) and **1** grammar (**Type 0**).
- There are **2** classes closed under complement: **Regular** and **Recursive** (and **CSL**); **CFL** and **RE** are **not**.

Power Comparisons

- For finite automata, **NFA = DFA** in power (determinism adds **0**).
- For pushdown automata, **NPDA > DPDA** in power (determinism loses **some** languages).
- For Turing machines, **NTM = DTM** in language power (determinism loses **0** languages).

State Counts

- "Divisible by k " needs **k** minimal-DFA states.
- " n -th from end is 1" needs **2^n** minimal-DFA states.
- "Ends with string w " needs **$|w|+1$** minimal-DFA states.
- Subset construction has worst-case blow-up of **2^n** states.

Canonical Languages

- **$a^n b^n$** is **CFL** but **not regular** (needs **1** count).
- **$a^n b^n c^n$** is **CSL** but **not CFL** (needs **2** counts).
- **ww** is **not CFL** (cannot match two halves with one stack).
- **a^p (p prime)** and **$a^{(n^2)}$** are **not regular and not CFL**.
- **$w w^R$** (palindrome) is **CFL** accepted by **NPDA** only.

Complexity of Algorithms

- **CYK** membership runs in **$O(n^3)$** time.
- DFA **membership** simulation runs in **$O(|w|)$** time.
- **Table-filling** minimization runs in **$O(n^2)$** ; **Hopcroft** runs in **$O(n \log n)$** .
- **NFA to DFA** subset construction is worst-case **$O(2^n)$** in states.

Decidability Quick Table

- **Equivalence** is decidable for **regular**, undecidable for **CFL/TM**.
- **Ambiguity** of CFGs is **undecidable**.
- **Emptiness** is decidable for **regular and CFL**, undecidable for **TM**.
- **Membership** is decidable for **regular, CFL, CSL/recursive** but only **semi-decidable** for **RE**.

13. Extended One-Liners (Atomic Coverage)

Strings and Languages (deep)

- The set of all strings of length exactly k over a c -symbol alphabet has c^k members.
- The number of strings of length $\leq n$ over a c -symbol alphabet is $(c^{n+1} - 1)/(c - 1)$.
- A **palindrome** reads the same in **2** directions: forwards and **backwards** ($w = w^R$).
- The **transpose/reversal** of **epsilon** is **epsilon**, the **1** fixed point of reversal among short strings.
- **Concatenation** of **0** copies of any string equals **epsilon**.
- A language is **finite** if it has a **finite** number of strings, else it is **infinite**.
- Every **finite** language is **regular**, requiring **0** exceptions.
- The **empty language** ϕ is **regular**, accepted by a DFA with **no** reachable final state.

DFA/NFA (deep)

- A DFA with n states accepting a non-empty language accepts some string of length $< n$.
- If a DFA accepts an infinite language, it accepts a string of length between n and $2n-1$ (pumping).
- A **complete** DFA defines **exactly 1** transition for **every** (state, symbol) pair.
- The **cross-product** of an m -state and an n -state DFA has up to $m \cdot n$ states for **intersection** or **union**.
- An NFA with n states has at most 2^n reachable subsets during conversion.
- The minimal DFA for Σ^* has exactly **1** state, which is **final**.
- The minimal DFA for ϕ has exactly **1** state, which is **non-final** (a dead state).
- A DFA recognising "**even length**" needs exactly **2** states.

Regular Expressions (deep)

- The RE $(a+b)^*$ denotes Σ^* over $\{a,b\}$, accepting **all** strings.
- The RE a^*b^* denotes strings with all **a**s before all **b**s, totalling **1** ordering constraint.
- The RE $(a+b)^*abb$ denotes strings ending in **abb**, needing a DFA of **4** states.
- The RE $(00+11)^*$ denotes even-length strings of **matched pairs**.
- **Two** regular expressions are **equivalent** if they denote the **same** language.

Closure recap (deep)

- **Regular** languages are closed under **quotient** with any language, giving **1** more closed operation.
- **Regular** languages are closed under **substitution** by regular languages.
- **CFLs** are closed under **substitution** and **homomorphism**, but **not inverse** of all maps... actually CFLs ARE closed under **inverse homomorphism**.
- **CFLs** are closed under **union with a CFL** but not **intersection with a CFL**.
- **Recursive** languages are closed under **set difference**; **RE** languages are **not**.

Pumping recap (deep)

- The pumping length p can be taken as the number of **states** in a DFA for L .
- Pumping **down** (taking $i=0$) removes y entirely, a valid case of $xy^i z$ in L .
- The CFL pumping length relates to the grammar's **variable** count and the $2^{|V|}$ bound.
- **Ogden's lemma** is a stronger version of the **CFL pumping lemma** using **marked** positions.

CFG/PDA recap (deep)

- A **linear grammar** has at most **1** variable per RHS but it may sit in the **middle**.
- $\{a^n b^n\}$ has a **linear** grammar but no **regular** one.
- Every **regular** language is a **CFL**, requiring **0** counterexamples.
- A **DCFL** is recognised by a **DPDA** and is closed under **complement**.
- The **CYK** table for a string of length **n** has $n(n+1)/2$ cells.
- A PDA accepting by **empty stack** can be converted to one accepting by **final state** and vice versa.

TM/Computability recap (deep)

- A **decider** halts on **100%** of inputs; a **recogniser** may halt on only the **accepted** ones.
- The set of **Turing machines** is **countably infinite**, fewer than the **uncountable** set of languages.
- This counting gap proves at least **1** language is **not RE**.
- **A_TM** reduces to **HALT_TM**, both being **RE but not recursive**.
- The complement of a **recursive** language is always **recursive**.
- An **LBA** is a TM whose tape is bounded to **O(n)** (linear in input length).
- Whether an **LBA** accepts the **empty** language is **undecidable**.
- **Context-sensitive** languages are closed under **complement** (Immerman-Szelepcsényi).

Decidability recap (deep)

- "Does a DFA accept a **finite** language?" is **decidable**.
- "Are two **CFGs** equivalent?" is **undecidable**.
- "Is a **CFL** regular?" is **undecidable**.
- "Is $L(M) = \phi$ for a TM M ?" is **undecidable** by **Rice's theorem**.
- "Does TM M have exactly **5** states?" is **decidable** (syntactic, not semantic).
- "Is a given CFG **ambiguous**?" is **undecidable**.
- The **PCP** over a **1**-symbol alphabet (single letter) is **decidable**, but general PCP is **undecidable**.

Hierarchy recap (deep)

- There are exactly **4** machine classes increasing in power: **FA < PDA < LBA < TM**.
- A **regular** language is recognised at all **4** levels of the hierarchy.
- **Type 1** grammars are **non-contracting** ($|LHS| \leq |RHS|$).
- **Type 0** grammars place **0** restrictions on production form (beyond a variable on the LHS).
- **Context-free** but not **regular**: requires **exactly 1** stack to recognise.
- **Context-sensitive** but not **context-free**: requires **linear** bounded memory.

Numeric/quick recall (deep)

- A **DFA** has **5** components, a **PDA** has **7**, and a **TM** has **7**.
- A **Moore/Mealy** machine has **6** components.
- A **CFG** has **4** components, a **regular expression** alphabet has **3** operators.
- **CNF** uses $2n-1$ steps; **GNF** uses n steps for a length- n string.
- **CYK** is $O(n^3)$; **Hopcroft** is $O(n \log n)$; subset construction is $O(2^n)$.
- The **Chomsky hierarchy** has **4** types, mapping to **4** machines and **4** language classes.

True/False recall (deep)

- "**NFA** is more powerful than **DFA**" is **FALSE** (they are equal).
- "**NPDA** is more powerful than **DPDA**" is **TRUE**.
- "**Regular** languages are closed under **intersection**" is **TRUE**.
- "**CFLs** are closed under **intersection**" is **FALSE**.
- "**CFLs** are closed under **complement**" is **FALSE**.
- "**RE** languages are closed under **complement**" is **FALSE**.
- "**Recursive** languages are closed under **complement**" is **TRUE**.
- "Every **context-free** language is **recursive**" is **TRUE**.
- "The **Halting problem** is **decidable**" is **FALSE**.
- " $a^n b^n c^n$ is **context-free**" is **FALSE** (it is context-sensitive).
- "The **pumping lemma** proves a language **IS** regular" is **FALSE** (it only proves non-regularity).
- "**Myhill-Nerode** is both necessary and sufficient for regularity" is **TRUE**.
- "CFG **equivalence** is **decidable**" is **FALSE**.
- "DFA **equivalence** is **decidable**" is **TRUE**.
- " L^* always contains **epsilon**" is **TRUE**.
- " $\phi^* = \phi$ " is **FALSE** (it equals $\{\epsilon\}$).

Mixed atomic facts (deep)

- A **transition diagram** marks the start state with **1** incoming arrow and final states with **double circles**.
- The **yield** of a parse tree is read **left to right** across the **leaves**.
- **Arden's theorem** is used in the **FA-to-RE** conversion.
- **Thompson's construction** is used in the **RE-to-NFA** conversion.
- A **universal language** L_u is **RE** but **not recursive**.
- **Diagonalization** is the technique proving the **Halting problem** undecidable.
- The **stack** of a PDA follows **LIFO** (last-in, first-out) discipline.
- A **2-stack PDA** is as powerful as a **Turing Machine**.
- A **queue automaton** (PDA with a queue) is also equivalent to a **Turing Machine**.
- A **counter machine** with **2** counters is equivalent to a **Turing Machine**.
- A **1-counter** automaton is strictly **weaker** than a **PDA**... it is a special case recognising some non-regular languages.
- The minimal DFA's state count equals the **Myhill-Nerode** index.
- Two DFAs are **equivalent** iff their **minimal** forms are **isomorphic**.
- A **regular** language always has **both** a left-linear and a right-linear grammar.
- The **reverse** of a regular language is **regular**.
- The **reverse** of a CFL is a **CFL**.
- A homomorphism replaces each **symbol** by a **string**, preserving **regularity** and **context-freeness**.
- **Closure** under **union** holds for **all** four major classes (Regular, CFL, Recursive, RE).
- **Closure** under **intersection** fails only for **CFL** among the four major classes.
- The **empty stack** acceptance mode ignores **final states** entirely.
- An **epsilon-NFA** with n states still converts to a DFA with at most 2^n states.
- **Kleene's theorem** and **Myhill-Nerode** both characterise **regular** languages.

Examiner-favourite numeric facts

- A binary-divisible-by-**3** DFA needs exactly **3** states.
- A "**contains 101**" DFA needs exactly **4** states (minimal).

- A "**third symbol from start is a**" DFA needs a constant number of states, around **5** including dead.
- The number of **distinct** DFAs (up to renaming) for a language is **1** minimal one.
- A **2-state** DFA can recognise "**ends in a**" over **{a,b}**.
- The **Cook-Levin** theorem dates to **1971**; Turing's halting proof to **1936**.
- **SAT** was the **1st** problem proven **NP-complete**.
- There are **3** resource classes commonly compared: **P**, **NP**, and **NP-complete**.

Final consolidation

- The **4** questions most asked in CIL MT TOC: **class identification**, **closure true/false**, **minimal-DFA counting**, and **decidability**.
- **Regular** = **finite memory**; **CFL** = **stack memory**; **CSL** = **linear memory**; **RE** = **unbounded memory**.
- The **single most-tested** contrast: **equivalence** is decidable for **regular** but not for **CFL**.
- The **second most-tested** contrast: **NFA=DFA** but **NPDA>DPDA** in power.
- The **third most-tested** contrast: **Recursive** closed under complement, **RE** not.
- **aⁿ bⁿ** (CFL), **aⁿ bⁿ cⁿ** (CSL), **ww** (non-CFL): the **3** canonical separating languages.
- A **TM** is the **1** model defining **algorithmic** computability via the **Church-Turing** thesis.
- Memorise the **4x** master table: **class -> machine -> grammar -> closure -> decidability** for full marks.

14. Additional Atomic One-Liners (Set 2)

Foundations extended

- An **alphabet** must exclude **0** infinite-size possibilities since it is by definition **finite**.
- The number of strings of length exactly **2** over $\{a,b\}$ is **4**: **aa, ab, ba, bb**.
- The number of strings of length exactly **3** over $\{0,1\}$ is **8** (2^3).
- The concatenation operation is **closed** on Σ^* , giving exactly **1** resulting string per pair.
- The **prefix closure** of a language adds all **prefixes** of its members.
- A **proper substring** of **w** excludes **2** cases: **w** itself and (commonly) **epsilon**.
- $(w^R)^R$ equals exactly **1** string: **w** (double reversal is identity).
- The **power** $a^0 b^0$ equals **epsilon**, of length **0**.
- A language over a **1**-symbol (unary) alphabet is called a **unary language**.
- Every **unary** context-free language is in fact **regular**, a **1**-of-a-kind collapse.

DFA extended

- A DFA accepting **exactly** the string **ab** needs **4** states including a **dead** state.
- The number of **final** states in a DFA can be **0**, meaning it accepts **phi**.
- A DFA with **all** states final still rejects **0** strings only if it is **complete** with no dead state.
- A DFA's language is **infinite** iff its transition graph has a **reachable** cycle leading to a **final** state.
- The **product** DFA for **difference** $L_1 - L_2$ accepts when DFA1 is **final** and DFA2 is **non-final**.
- A DFA recognising "**starts with a**" over $\{a,b\}$ needs **3** states (incl. dead).
- A DFA for "**contains an even number of 0s and odd number of 1s**" needs **4** states (2×2 product).
- The **empty** DFA (no states) is typically **disallowed**; a DFA needs at least **1** state.
- The **trap** state contributes **0** accepted strings.
- Counting the minimal DFA for "**divisible by 6**" gives **6** states.

NFA / conversion extended

- An NFA may have **multiple** start states in some textbook variants, normalised to **1** via **epsilon**.
- The subset construction's reachable-states count is $\leq 2^n$ but often far **fewer**.
- An NFA accepting "**ends with ab**" needs **3** states; its DFA needs **3** states too (no blow-up here).
- The **worst-case** blow-up example is "**n-th symbol from end**", needing 2^n DFA states.
- Converting an **epsilon-NFA** first computes **epsilon-closure** of **1** start state.
- A DFA state in subset construction is **dead** if it equals the **empty set**.

Minimization extended

- Two final states are always **indistinguishable** by the **empty** string, needing a longer **witness**.
- The minimization **partition refinement** starts with **2** groups: **final** and **non-final**.
- Each refinement step can only **increase** or keep the number of **partitions**.
- Minimization terminates when **no** partition splits further, giving the **minimal** DFA.
- The minimal DFA is **unique** up to **state renaming (isomorphism)**.

Regular expression extended

- The RE **a+** equals **aa***, denoting **1** or more **as**.

- The RE **a?** (optional) equals **epsilon + a**, denoting **0 or 1 a**.
- The RE **(ab)*** denotes strings that are **repetitions** of the block **ab**.
- The RE **a*(b a*)*** can describe strings over **{a,b}** with **any** arrangement.
- **Two** REs denoting the same language are **equivalent**, decided via **minimal DFA** comparison.
- The RE for "**even number of a's**" over **{a}** is **(aa)***.
- The RE for "**strings not containing 11**" uses **(0+10)*(1+epsilon)**.

Closure extended

- **Regular** is closed under **init/prefix**, **suffix**, and **cycle** operations.
- **Regular** is closed under **max** and **min** operations on languages.
- **CFL** is closed under **cyclic shift**.
- **CFL $\cap$ CFL** may be **non-CFL**, shown by **aⁿ bⁿ cⁿ**.
- **Recursive $\cap$ Recursive** is always **recursive**.
- **RE $\cap$ RE** is always **RE**; **RE complement** may **not** be RE.

Pumping extended

- For a regular language, the **pump** repeats **1** block **y** any number of times.
- For a CFL, the **pump** repeats **2** blocks **v** and **x** in lockstep.
- The pumping lemma is a **game**: the adversary picks **p**, you pick **w**, adversary splits, you pick **i**.
- To break regularity you choose **1** value of **i** (often **i=2** or **i=0**) putting the string **outside L**.
- **aⁿ bⁿ** fails regular pumping because pumping **as** breaks the **equality** of counts.
- **aⁿ bⁿ cⁿ** fails CFL pumping because **vw^x** cannot span all **three** blocks.

CFG / PDA extended

- **Leftmost** and **rightmost** derivations of an **unambiguous** grammar give the **same** parse tree.
- A **derivation step** $\Rightarrow^*$ denotes **zero or more** rewriting steps.
- **Useless symbol** removal can shrink a grammar to **0** productions if **L** is empty.
- A **nullable** variable derives **epsilon** in **≥ 1** step.
- **CNF** forbids **epsilon**-productions except possibly **S $\rightarrow$ epsilon**.
- A **DPDA** has **at most 1** applicable move per configuration.
- The **2-stack** PDA jumps to **full** Turing power.
- **NPDA** accepts **{ww^R}**; **DPDA** cannot, a **1-language** power gap demonstration.
- A **PDA** can be built from a **CNF** grammar with a **single** state.
- The **height** of a PDA stack can be **unbounded**, unlike the **0** memory of an FA.

CFL decision extended

- **CYK** fills a triangular table of **n(n+1)/2** cells in **O(n³)** time.
- CFL **membership** is decidable; CFL **equivalence** is **undecidable**.
- CFL **emptiness** reduces to checking if **S** is a **generating** symbol.
- CFL **finiteness** reduces to **cycle detection** in a cleaned grammar's graph.
- "Is a CFL equal to **Sigma***?" is **undecidable**.

TM extended

- A TM tape extends to **infinity** in at least **1** direction.

- A **multi-tape** TM with **k** tapes simulates in $O(t^2)$ on **1** tape.
- A **non-deterministic** TM with time **t** simulates deterministically in $2^{O(t)}$ time.
- The **UTM** demonstrates a **single** machine can run **any** program.
- An **enumerator** for a **recursive** language outputs strings in **canonical** order.
- A TM that **halts** on all inputs decides a **recursive** language.
- The **busy beaver** function is **non-computable**, growing faster than any **computable** function.
- A **linear bounded automaton** uses $O(n)$ tape cells.

Computability extended

- **Recursive** $\subset$ **RE** with the gap witnessed by **A_TM**.
- A language and its complement both **RE** implies the language is **recursive** (Post).
- **Co-RE** is the class of **complements** of RE languages.
- **A_TM complement** is in **co-RE** but **not** in **RE**.
- The class **RE** $\cap$ **co-RE** equals exactly the **recursive** languages.

Decidability extended

- "Does a TM accept ≥ 1 string?" is **undecidable** (non-emptiness, by Rice).
- "Does a TM accept a **regular** language?" is **undecidable** (Rice).
- "Does a TM accept a **context-free** language?" is **undecidable** (Rice).
- "Does a TM run for **>100** steps on input w?" is **decidable** (just simulate).
- "Is the language of a DFA **infinite**?" is **decidable**.
- "Do two **TMs** accept the same language?" is **undecidable**.
- "Is a TM's language **empty**?" is **undecidable** (**E_TM**).
- "Is **w** accepted by TM **M** within **k** steps?" is **decidable**.
- **Rice's theorem** covers **all** non-trivial **semantic** properties, totalling **0** decidable ones among them.

Complexity extended

- **P** is contained in **NP**, which is contained in **PSPACE**.
- **NP** problems have **polynomial-size** certificates verifiable in **polynomial** time.
- A problem is **NP-complete** if it is in **NP** and **NP-hard**.
- **TSP** (decision version) is **NP-complete**; the **optimization** version is **NP-hard**.
- **P = NP** is **unresolved** and worth a **1-million-dollar** prize.
- **Cook-Levin** reduced every NP problem to **SAT**.
- **Reductions** used in NP-completeness run in **polynomial** time.

Master recall (Set 2)

- The **4** closure facts to memorise: Regular **all**, CFL **not** $\cap$ /complement, Recursive **all incl complement**, RE **not complement**.
- The **3** power facts: **NFA=DFA**, **NPDA>DPDA**, **NTM=DTM**.
- The **4** machine-tuple sizes: **DFA=5**, **Moore/Mealy=6**, **PDA=7**, **TM=7**.
- The **2** step-count formulas: **CNF=2n-1**, **GNF=n**.
- The **3** algorithm complexities: **CYK=O(n^3)**, **Hopcroft=O(n log n)**, **subset=O(2^n)**.
- The **3** undecidable CFL problems: **equivalence**, **ambiguity**, **totality (L=Sigma*)**.
- The **2** historical dates: **Halting 1936**, **Cook-Levin 1971**.
- The **4** canonical non-regular/non-CFL languages: **a^n b^n**, **a^n b^n c^n**, **ww**, **a^p (prime)**.

- The **1** theorem that is necessary AND sufficient for regularity: **Myhill-Nerode**.
- The **1** model defining computability: the **Turing Machine** via **Church-Turing**.
- The **4**-row master table: each class maps to a **machine**, **grammar**, **closure profile**, and **decidability profile**.
- **Regular** $\subset$ **DCFL** $\subset$ **CFL** $\subset$ **CSL** $\subset$ **Recursive** $\subset$ **RE** captures **6** nested classes.
- **DCFL** sits strictly between **Regular** and **CFL**, recognised by a **DPDA**.
- The **empty stack** and **final state** PDA modes coincide for **NPDA** but **not** for **DPDA**.
- **Greibach Normal Form** removes **left recursion** as a side effect.
- A grammar with production **S $\rightarrow$ aSb** alone generates **$\{a^n b^n : n \geq 1\}$** ... adding **S $\rightarrow$ ab** or **epsilon** sets the base.
- The **language** accepted by a minimal DFA and its source DFA are **identical**.

15. Final Atomic One-Liners (Set 3)

- A **symbol** is the **1** smallest indivisible unit of an alphabet.
- The **Kleene closure** of $\{a\}$ gives $\{\epsilon, a, aa, aaa, \dots\}$, an **infinite** set.
- A language with exactly **0** strings is **phi**; with exactly **1** string **epsilon** is $\{\epsilon\}$.
- A **DFA** reading **epsilon** stays in **1** place: the **start state**.
- An NFA's **delta[^]** returns a **set** of states, possibly the **empty** set.
- The **epsilon-closure** of a set is the **union** of closures of its **members**.
- A **minimal** DFA has **0** redundant (equivalent) states.
- **Moore** machines output even before reading **1** symbol (the start-state output).
- A **regular expression** and a **DFA** describe the **same** regular language by **Kleene's theorem**.
- The **Arden** solution QP^* is unique iff **P** lacks **epsilon**.
- **Regular** languages are closed under **symmetric difference** (a Boolean operation).
- The **pumping** constant equals the DFA's **state** count.
- **Myhill-Nerode** classes partition Σ^* into **finitely** many groups iff **L** is regular.
- A **CFG** can be ambiguous while its **language** has an **unambiguous** grammar.
- **Inherent ambiguity** means **every** grammar for the language is **ambiguous**.
- **CNF** restricts RHS to **2** variables or **1** terminal.
- **GNF** restricts RHS to **1** leading terminal followed by **variables**.
- A **PDA** stack uses **LIFO** order with initial symbol **Z0**.
- **NPDA** = **CFL** and **DPDA** = **DCFL**, with $DCFL \subset CFL$.
- $CFL \cap Regular$ stays **CFL**, the **1** intersection that is safe.
- **CYK** requires the grammar in **CNF** and runs in $O(n^3)$.
- A **TM** accepts, rejects, or **loops**, totalling **3** possible fates.
- **Recursive** = **decidable** = TM that **always halts**.
- **RE** = **semi-decidable** = TM that **halts on acceptance**.
- **Recursive** $\subset$ **RE**, with **A_TM** in **RE** but not **recursive**.
- The **Halting problem** reduces to most **undecidable** TM problems.
- **Rice's theorem** makes all **non-trivial language** properties **undecidable**.
- **PCP** is **undecidable** but its **unary** restriction is **decidable**.
- The **universal language** L_u is **RE**, its **complement** is **not RE**.
- **P** is **deterministic polynomial**; **NP** is **non-deterministic polynomial**.
- **SAT** is **NP-complete** by **Cook-Levin (1971)**.
- **NP-hard** need not be **in NP**, e.g. **Halting problem**.
- The hierarchy of memory: **none** < **stack** < **linear-tape** < **unbounded-tape**, matching **FA, PDA, LBA, TM**.
- A **2-counter machine** matches **Turing** power, surprising for just **2** counters.
- **Context-sensitive** languages are closed under **complement** (Immerman-Szelepcsenyi, 1987).
- **Regular** languages have a **unique** minimal DFA but **many** equivalent NFAs.
- The **reversal** of a regular language is recognised by **reversing** all DFA edges and swapping **start/final**.
- **Thompson's construction** adds **epsilon**-edges for **union, concatenation, star**.
- A **dead state** is needed to **complete** a partial DFA before **complementation**.
- The **3** most common CIL MT TOC question types: **construct, classify, closure/decidability true-false**.
- $a^n b^n$ needs **1** stack; $a^n b^n c^n$ needs **2** comparisons hence an **LBA**.
- Σ^* is **countable**; the set of all **languages** is **uncountable**, forcing some **non-RE** languages.
- The **minimal** number of states to recognise **phi** is **1** (a single dead state).

- The **minimal** number of states to recognise **Sigma*** is **1** (a single final state).
- The **Chomsky** mapping to memorise: **Type 3=FA, Type 2=PDA, Type 1=LBA, Type 0=TM.**